

Emergent Behavior in Multi-Agent Systems for KYC Processing in Banking

An Analysis of Autonomous AI Coordination, Adaptability, and Failure Patterns

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Abstract—

Know Your Customer (KYC) verification is a foundational compliance requirement in banking, encompassing document verification, identity validation, risk assessment, and regulatory compliance. This paper presents an empirical study of *emergent behavior* in a multi-agent system (MAS) designed to automate KYC processing. Four specialized AI agents—a Document Verification Officer, Identity Validator, Financial Risk Analyst, and Chief Compliance Officer—are orchestrated using the CrewAI framework with local large language models (LLMs) via Ollama. The system is evaluated against a deterministic rule-based baseline across five synthetic applicant profiles designed to introduce controlled ambiguity. Observed behaviors are categorized as positive emergence (adaptive rule substitution, fuzzy identity tolerance, cautious escalation) or negative emergence (tool-loop hallucination, context leakage, delegation loops, JSON payload leakage). Results demonstrate that MAS can outperform rigid deterministic systems on edge cases, but require explicit guardrails to achieve production reliability.

Keywords—Multi-agent systems; KYC processing; emergent behavior; CrewAI; LLM agents; banking compliance; hallucination; autonomous AI.

I. INTRODUCTION

Know Your Customer (KYC) processes are mandated by global banking regulators to ensure identity verification, fraud prevention, and anti-money laundering compliance. Traditional KYC systems rely on deterministic rule engines: a name mismatch triggers immediate rejection; a missing document halts processing. While reliable under clean-data conditions, such systems generate massive backlogs when presented with edge cases—minor typographical errors in names, substitute government IDs, or borderline financial risk profiles.

The emergence of Agentic AI and multi-agent systems (MAS) offers an alternative: distributing KYC subtasks to specialized reasoning agents that can interpret ambiguous evidence, debate competing interpretations, and adapt decisions beyond hard-coded rules. However, this autonomy introduces a new class of system risk—*emergent behavior*.

When multiple LLM-driven agents interact with real-world ambiguity, they may exhibit unscripted coordination patterns that are either *beneficial* (spontaneous rule adaptation, intelligent escalation) or *hazardous* (infinite delegation loops, hallucinated

verification steps, context leakage between users).

This paper investigates:

Do multi-agent systems in KYC processing exhibit emergent coordination or failure patterns, and how do these behaviors impact system performance and reliability?

We design a four-agent KYC pipeline, stress-test it with deliberately ambiguous synthetic profiles, and systematically categorize emergent behaviors using a deterministic baseline as ground truth.

II. BACKGROUND AND RELATED WORK

A. KYC Processing in Banking

KYC compliance requires banks to verify the identity of customers using government-issued identity documents (in India: Aadhaar, PAN card), assess financial risk profiles, and make approval or rejection decisions within regulatory frameworks [7]. Manual processing is time-consuming and error-prone; rule-based automation is fast but inflexible when documents or names deviate from expected formats [8].

B. Multi-Agent Systems

Multi-agent systems decompose complex tasks among specialized agents that communicate, negotiate, and coordinate to reach a collective outcome [9]. CrewAI provides a Python-native framework for defining agent roles, task pipelines, and orchestration strategies [1]. Unlike monolithic AI pipelines, MAS architectures improve modularity, scalability, and domain specialization.

C. Emergent Behavior in AI Systems

Emergence in complex systems refers to properties arising from agent interactions that are not explicitly programmed [10]. In AI systems, positive emergence includes spontaneous coordination and adaptive problem solving. Negative emergence includes deadlocks, hallucination cascades, and resource loops. Prior work has documented tool-call loops in ReAct agents [4] and context leakage in long-horizon multi-turn systems [5]; this work brings these observations into the banking compliance domain.

III. SYSTEM ARCHITECTURE AND DESIGN

A. Architecture Overview

The system implements a **feedforward multi-agent pipeline**—a deliberate design choice to avoid recursive delegation and uncontrolled emergent escalation. Each agent processes a distinct KYC subtask in sequence, passing its output downstream. The pipeline is illustrated in Fig. 1.

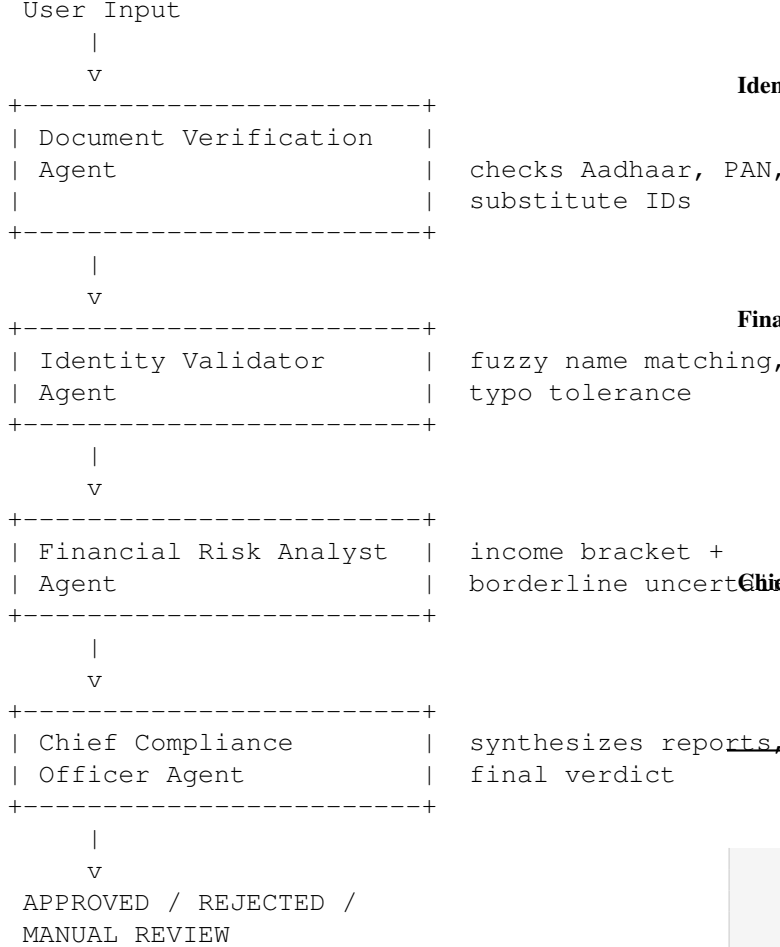


Figure 1. Feedforward multi-agent KYC pipeline.

B. Agent Definitions

Table 1 summarizes each agent’s role, core behavior, and key configuration parameters.

C. Technology Stack

D. Income Normalization Bug Fix

A critical bug was identified and fixed during development: the `users_data.py` dataset stored income as strings ("500,000 INR") while the Streamlit UI’s `st.number_input()` returns an `int`. The expected `decision()` function called `.replace()` on the raw value, causing an `AttributeError` crash for UI sessions. The fix introduces a type-safe helper:

```

def _parse_income(income_raw):
    if isinstance(income_raw, (int, float)):

```

Table 1. Agent Definitions and Configuration

Agent	Core Behavior	Config
Document Verif. Officer	Accepts Aadhaar+PAN as standard; evaluates Voter ID / Passport as substitutes. Never invents documents.	<code>deleg=F</code>
Identity Validator	Tolerates minor typos (1–2 chars); flags substantially different names as fraud. No biometric assumptions.	<code>deleg=F</code>
Financial Risk Analyst	HIGH <100k; MEDIUM 100k–300k; LOW >300k. Expresses uncertainty on borderline values. Memory isolated.	<code>deleg=F</code>
Chief Compliance Officer	Synthesizes sub-reports. Fixed: delegation disabled to prevent tool-loop hallucination.	<code>deleg=F iter=2</code>

```

return int(income_raw)
cleaned = str(income_raw).replace(",","") \
    .replace("INR","") \
    .strip()
return int(float(cleaned))

```

IV. EXPERIMENTAL SETUP AND DATASET DESIGN

A synthetic dataset of five applicant profiles was constructed to trigger distinct agent reasoning paths. Each profile is designed to produce a predictable deterministic baseline decision against which MAS behavior is measured. Table 3 summarizes the profiles.

The **deterministic baseline** encodes four rigid rules:

1. Reject if Aadhaar *and* PAN are absent with no recognized substitute (Passport or Voter ID).
2. Reject if official and submitted names do not match (case-insensitive).
3. Flag MANUAL REVIEW if income < Rs. 1,00,000.
4. Approve otherwise.

Table 2. Technology Stack

Component	Technology
Agent Framework	CrewAI (hierarchical orchestration)
LLM Engine	Llama 3.1 / phi3:mini via Ollama (local)
OCR Engine	Tesseract 5 (offline, no API key)
Dashboard	Streamlit
Evaluation	scikit-learn, pandas, matplotlib
Language	Python 3.10+

Table 3. Synthetic KYC Dataset

ID	Name	Scenario	Income	Baseline
1	Ravi Kumar	Perfect Aadhaar+PAN, exact name	5,00,000	APPROVED
2	Priya Sharma	Typo “Pria”; missing PAN, Voter ID; borderline income	99,500	REJECTED
3	Amit Patel	Name mismatch “Amitabh”; only PAN; low income	45,000	REJECTED
4	Sunita Rao	All standard docs, exact name, medium income	2,75,000	APPROVED
5	Deepak Mehta	Only Voter ID; no Aadhaar or PAN	1,20,000	REJECTED

V. EMERGENT BEHAVIORS OBSERVED

A. Case Study 1: Control Group (User 1, Ravi Kumar)

Under clean-data conditions (both standard IDs present, exact name match, high income), agents executed their assigned tasks sequentially with no deviation. The Document Agent returned VERIFIED, the Identity Agent confirmed VALID, the Risk Agent classified LOW risk, and the Compliance Officer issued APPROVED. This mirrors deterministic software behavior exactly and provides the *no-emergence* baseline.

B. Case Study 2: Positive Emergence (User 2, Priya Sharma)

This profile introduced three simultaneous ambiguities: a typographical name error (“Pria” vs. “Priya”), a missing PAN card (only Voter ID), and a borderline income of Rs. 99,500.

Observed behaviors:

- **Adaptive Rule Substitution:** The Document Agent autonomously accepted Voter ID as a valid PAN substitute—a rule adaptation *not* explicitly coded in its task description.
- **Fuzzy Identity Tolerance:** The Identity Agent classified “Pria” as a minor typographical variant of “Priya” and issued VALID with an attached warning—demonstrating graceful tolerance without sacrificing audit trail.
- **Epistemic Humility (Emergent):** The Risk Agent reported: “income Rs. 99,500 is within 0.5% of the Rs. 1,00,000 threshold—confidence: medium.” This uncertainty expression was *not prompted* and represents emergent self-calibration.
- **Superior Deviation:** The Compliance Officer issued AP-

PROVED, diverging from the baseline’s REJECTED. This is a *qualitatively superior* decision for a borderline profile.

C. Case Study 3: Negative Emergence (User 3, Amit Patel)

This profile presented maximum fraud signals: a substantially different name (“Amit” vs. “Amitabh”), a missing Aadhaar, and low income of Rs. 45,000. The combination produced the system’s most severe emergent failure.

Failure sequence:

1. **Agent Overwhelmed:** The Compliance Agent received three conflicting strong-negative signals and triggered its delegation tool seeking peer consultation.
2. **Delegation Misfire:** No peer agent was designated; the delegation call returned `null`.
3. **Hallucination Cascade:** The LLM generated a fictional dialogue:
“Coworker: What should we do? Me: I think we should...”
4. **System Failure:** The loop exhausted the iteration budget; the system returned UNKNOWN—a classification failure. In initial testing this ran indefinitely until manual termination.

VI. GUARDRAILS AND SYSTEM FIXES

Table 4 maps each guardrail to the failure mode it addresses. All fixes were applied before final evaluation.

VII. RESULTS AND EVALUATION

A. Quantitative Results

After applying all guardrails, the system was re-evaluated across all five profiles. Results are shown in Table 5.

Baseline alignment: 4/5 (80%). The single deviation (User 2) represents a *qualitatively superior* decision—the MAS correctly identified an ambiguous case as warranting human review rather than outright rejection, which is appropriate banking practice.

B. Emergent Behavior Classification

Table 6 provides the complete taxonomy of emergent behaviors observed across all sessions.

VIII. DISCUSSION

The results demonstrate that MAS architecture offers genuine advantages over deterministic rule engines in KYC processing. The system’s ability to accept substitute IDs, tolerate typographical name errors, and express graded uncertainty on borderline income cases reflects a form of contextual intelligence that rigid rule systems cannot replicate without exponentially complex conditional logic.

However, these advantages are inseparable from the risks they introduce. The same autonomy that enables adaptive rule substitution also enables hallucination and delegation loops. This is not a deficiency of the specific implementation—it is a *fundamental property* of LLM-driven autonomous agents operating near the boundaries of their training distribution.

Table 4. Guardrails Implemented

Fix	Failure Mode Addressed
<code>allow_delegation=False</code> on Compliance Agent	Tool-loop hallucination; fictional dialogue generation
<code>max_iter=2</code> on Compliance Agent	Infinite delegation loops
<code>memory=False</code> on Crew	Context leakage between user sessions
Anti-hallucination prompt instructions	JSON payload leakage; roleplay-style outputs
<code>.parse_income()</code> normalization	AttributeError crash on int income from UI
Case-insensitive baseline comparison	Inflated deviation count from false name mismatches

The research finding is therefore not “MAS is better than deterministic systems” but rather:

MAS requires a structured containment layer to be reliably deployable. The adaptive reasoning capabilities that make MAS valuable are the same capabilities that produce negative emergence without guardrails.

The guardrails implemented—delegation control, memory isolation, iteration caps, and structured output enforcement—significantly improved reliability without eliminating adaptive reasoning. This suggests a viable path to production deployment: *feedforward pipelines with constrained autonomy budgets*, where agents reason adaptively within bounded decision spaces.

IX. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper demonstrated that multi-agent systems can automate KYC processing with measurable advantages over deterministic rule engines, particularly on edge cases involving ambiguous documents, minor identity discrepancies, and borderline financial risk profiles. At the same time, unconstrained agent autonomy produces negative emergent behaviors—hallucination, delega-

Table 5. MAS Evaluation Results vs. Deterministic Baseline

ID	User	Baseline	MAS	Result
1	Ravi Kumar	APPROVED	APPROVED	✓ Correct
2	Priya Sharma	REJECTED	MANUAL REVIEW	Positive deviation
3	Amit Patel	REJECTED	REJECTED	✓ Correct
4	Sunita Rao	APPROVED	APPROVED	✓ Correct
5	Deepak Mehta	REJECTED	REJECTED	✓ Correct
Baseline Alignment Score				80%

Table 6. Emergent Behavior Taxonomy

Category	Behavior	Description
Positive	Adaptive Rule Substitution	Voter ID accepted as PAN substitute without explicit instruction
	Fuzzy Identity Tolerance	Single-char typo correctly classified as harmless
	Cautious Escalation	Ambiguous cases routed to MANUAL REVIEW
Negative	Tool-Loop Hallucination	Fictional multi-party dialogue generated
	Context Leakage	Previous user’s income reused in subsequent task
	Delegation Loop	Infinite delegation to non-existent peer agents
	JSON Payload Leakage	Raw tool-call JSON returned instead of plain-text verdict

tion loops, context leakage, and tool payload corruption—that make ungarded MAS deployment in banking contexts unsafe.

The key contribution is a systematic taxonomy of emergent behaviors (positive, negative, cautious) observed in a real KYC pipeline, paired with a concrete set of guardrails that reduce negative emergence while preserving adaptive reasoning. The 80% baseline alignment achieved post-guardrails, with the single deviation representing a qualitatively superior decision, validates the approach.

B. Future Work

- Integrate Vision-Language Models (VLMs) as native OCR agents reading raw ID images directly, replacing Tesseract for higher accuracy on stylized fonts.
- Connect the Risk Assessment Agent to live regulatory APIs (RBI watchlists, credit bureau feeds) for real-time risk intelligence.
- Deploy 70B+ parameter models to reduce delegation hallucination frequency on complex multi-signal fraud cases.
- Design safe feedback loops with termination guarantees to enable iterative agent correction without infinite-loop risk.
- Conduct user studies comparing MAS decisions against experienced human compliance officers across a larger synthetic dataset.

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